

AREC 261 · Module 1 Test — SAMPLE

Dataset for this test: Saskatchewan RM Crop Yields. You are provided two files: the wide file `rm_yields_1990plus.csv` (one column per crop) and the long file `rm_yields_1990_2025.csv` (crop name in its own Crop column, for PivotTables). Lentils are recorded in **pounds** per acre; every other crop is in bushels.

Submission. Submit one `.xlsx` workbook with a separate, clearly labelled worksheet for each question. For each question, 20% of the mark is for presentation (labelled cells, sensible formats, a reader can follow the sheet) and 80% for correctness.

Question 1

Enter this table into a blank sheet.

Field	Acres	Yield (bu/ac)
Kestrel	120	41.2
Meadowvale	240	38.6
Nightjar	95	44.8
Home	310	36.1
Rented	175	39.9

Off to the right of the table, in a column with nothing else in it, put the canola price **\$14.20** in a single labelled cell.

- (a) Add a **Revenue (\$/ac)** column: each field's yield times the price cell. Write the formula once and fill it down. Format as currency with two decimals.
- (b) Add a **Total revenue (\$)** column: revenue per acre times that field's acres. Format with a thousands separator and no decimals.
- (c) Under the table, use SUM to total the acres, the total bushels (yield × acres), and the total revenue.
- (d) In a spare column, repeat the (a) formula *without* the \$ signs and fill it down all five rows. Label the block clearly as the incorrect version.
- (e) Beside it, write one sentence saying what appears in those cells and why.

Question 2

Work with the **Canola** column for 2023 (copy the 2023 rows to a sheet of their own first).

- (a) Compute the mean and the median canola yield.
- (b) Compare the two values. What does the relationship between the mean and the median tell you about how yields were distributed across RMs that year, and about the low-yielding RMs in particular?
- (c) Compute the standard deviation and the interquartile range (IQR) of the same column.

(d) The IQR ignores the most extreme RMs while the standard deviation does not. Given what you found in (b), explain which of these two spread measures you would trust more to describe a “typical” amount of variation across RMs, and why.

(e) A classmate uses the mean to describe “a typical RM.” Do you think the mean or the median better represents a typical RM here? Justify your choice using your answers to (a)–(d).

Question 3

Use the full (all-years) wide file.

(a) Use AVERAGEIF to find the average canola yield in each of 2010, 2019, and 2021.

(b) Rank those three years by average yield, and identify which one stands out as unusually low.

(c) Take the lowest of those three years. Describe what you would check across the *other* crop columns to decide whether the low year was a broad event affecting all crops, rather than a canola-only result. (You do not need outside knowledge — describe the data check.)

(d) Your AVERAGEIF for 2021 returns a number even though several RMs have a blank canola cell that year. Explain what AVERAGEIF did with those blanks.

(e) Explain why filling those blanks with 0 before averaging would give a wrong answer.

Question 4

Use the **long** file.

(a) Build a PivotTable of average yield by crop for 2023 (Year in Filters = 2023, Crop on Rows, Yield as Average). Rank the crops.

(b) Look at the ranked averages. If one crop’s value looks out of line with the others, identify it and explain why it is so different.

(c) Explain what a reader would wrongly conclude if they took the ranking at face value.

(d) Add Unit to Filters and set it to bu/ac. Identify the highest and lowest crop now.

(e) Explain why the filtered comparison is the fair one.